Linear Equations

Algebra · Edexcel IGCSE Mathematics A (4MA1) Paper 1 · Year 10 · worksheet

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Name	Class	Date

Answer ALL questions.

Write your answers in the spaces provided. You must write down all the stages in your working.

Suggested time: 96 minutes · Total marks: 96

KEY RULES

Balance method	Do the same to both sides
Brackets	Expand first, then collect the x terms
Unknown on both sides	Move x terms to the side with more x
Fractions	Multiply every term by the common denominator
Check	Substitute your answer back in

WORKED EXAMPLE 1

[2 marks]

Solve $5x - 7 = 18$

Step 1	Add 7 to both sides	$5x = 25$
Step 2	Divide both sides by 5	$x = 5$

Answer: $x = 5$

WORKED EXAMPLE 2

[3 marks]

Solve $3(x + 4) = 2x + 15$

Step 1	Expand the bracket	$3x + 12 = 2x + 15$
Step 2	Subtract $2x$ from both sides	$x + 12 = 15$
Step 3	Subtract 12	$x = 3$

Answer: $x = 3$

WORKED EXAMPLE 3

[3 marks]

Solve $\frac{x+3}{2} = \frac{2x-1}{3}$

Step 1	Multiply both sides by 6	$3(x + 3) = 2(2x - 1)$
Step 2	Expand	$3x + 9 = 4x - 2$
Step 3	Collect the x terms	$11 = x$

Answer: $x = 11$

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1 Solve $7x - 3 = 2x + 12$

(Total for question = 2 marks)

Practice question written for this worksheet (not from a past paper)

2 Solve $4(2x - 1) = 3(x + 6)$

(Total for question = 2 marks)

Practice question written for this worksheet (not from a past paper)

3 Solve $\frac{6}{x-2} = 3$

(Total for question = 2 marks)

Practice question written for this worksheet (not from a past paper)

4 Solve $4x - 13 = 17 + 8x$

$x =$ .....

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5 Solve $\frac{3x-1}{4} - \frac{x+2}{3} = 1$

(Total for question = 3 marks)

Practice question written for this worksheet (not from a past paper)

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6 Sam is 4 years older than Tara. Their ages add up to 30.

Work out Sam's age.

(Total for question = 3 marks)

Practice question written for this worksheet (not from a past paper)

7 A bag has x sweets. Anna has 3 bags and 5 extra sweets.

Ben has 2 bags and 17 extra sweets. They have the same number of sweets.

Work out x.

(Total for question = 3 marks)

Practice question written for this worksheet (not from a past paper)

8 Solve $8y - 18 = 3(y + 3)$

Show clear algebraic working.

$y =$ .....

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9 Solve $6(3y + 5) = 39$
Show clear algebraic working.

$y =$ .....

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10 Solve $7x - 5 = 3x + 2$
Show your working clearly.

$x =$ .....

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11 Solve $3(2 - 4x) = 5 - 8x$
Show clear algebraic working.

$x =$ .....

(Total for question = 3 marks)

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12 Solve $4(5y - 1) = 3(6y + 7)$
Show clear algebraic working.

$y =$ .....

Edexcel IGCSE Mathematics A (4MA1) · Paper 3H · January 2015 · Question 6

13 Solve $7y - 6 = 2y + 8$

Show clear algebraic working.

$y =$ .....

Edexcel IGCSE Mathematics A (4MA1) · Paper 4H · June 2013 · Question 6

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14 Solve $3(2x + 5) = 4 - x$
Show clear algebraic working.

$x =$ .....

Edexcel IGCSE Mathematics A (4MA1) · Paper 3H · January 2016 · Question 8

15 Solve $3x + 16 = 1 - 2x$
Show clear algebraic working.

$x =$ .....

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16 **13** Solve $\frac{x+3}{4} - \frac{7-x}{5} = 4.3$

Show clear algebraic working.

$x =$ .....

(Total for Question 13 is 3 marks)

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17 Rectangle **A** has a width of x metres and a height of $(x + 2)$ metres.
Rectangle **B** has a width of $2x$ metres and a height of $4x$ metres.

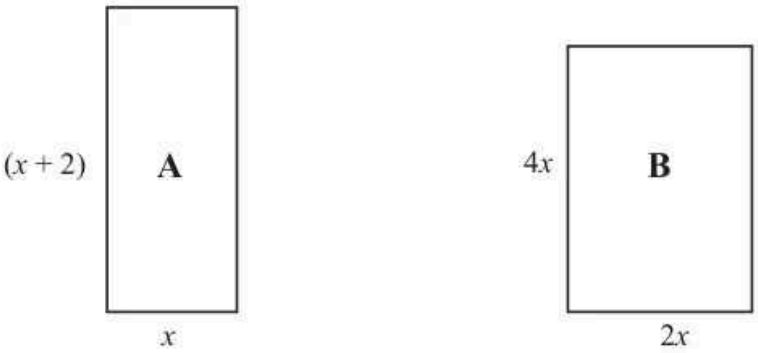


Diagram **NOT** accurately drawn

The perimeter of rectangle **A** is equal to the perimeter of rectangle **B**.

(i) Use this information to write down an equation in x.

.....

(ii) Find the value of x.

$x =$ .....

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18 The diagram shows rectangle $ABCD$

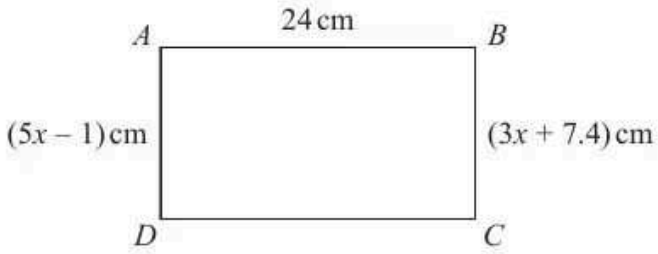


Diagram **NOT** accurately drawn

Work out the perimeter of the rectangle.
Show your working clearly.

..... cm

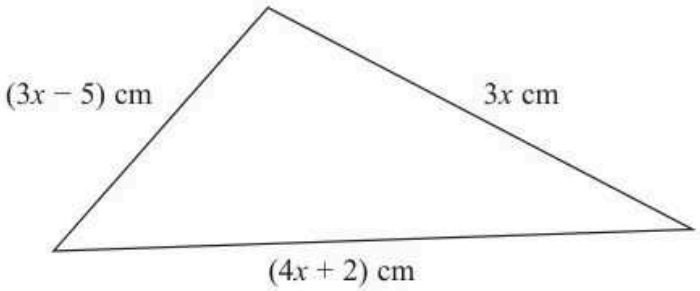
(Total for Question 4 is 4 marks)

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19 The diagram shows a triangle.

Diagram **NOT** accurately drawn



The lengths of the sides of the triangle are $3x$ cm, $(3x - 5)$ cm and $(4x + 2)$ cm.

The perimeter of the triangle is 62 cm.

Work out the value of x.
Show clear algebraic working.

$x =$ .....

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20 Solve $\frac{6x-1}{4} - \frac{5-2x}{2} = 1$

Show clear algebraic working.

$x =$ .....

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21 Solve $\frac{2x-1}{4} + \frac{x-1}{5} = 2$

$x =$ .....

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22 Solve $\frac{5-x}{2}-\frac{x-1}{3}=1$

Show clear algebraic working.

.....

Edexcel IGCSE Mathematics A (4MA1) · Paper 4HR · June 2017 · Question 14

23 Solve $\frac{}{5x-2}=\frac{}{6x+1}$

Show clear algebraic working.

$x =$.....

Edexcel IGCSE Mathematics A (4MA1) · Paper 4H · January 2013 · Question 14

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CHALLENGE QUESTIONS

24 (a) Solve $3(2x - 1) = 6$

Show clear algebraic working.

$x =$ .....

(3)

(b) Solve $\frac{2y+1}{3} = \frac{y-2}{4}$

Show clear algebraic working.

$y =$ .....

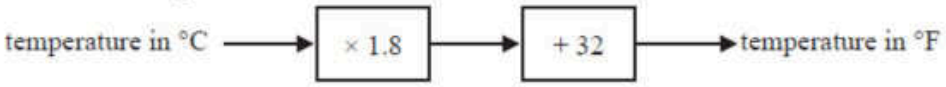
(4)

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25 Charlene is comparing cooking temperatures in degrees Celsius (°C) with cooking temperatures in degrees Fahrenheit (°F)

She finds this rule to change from °C to °F



(a) Change a temperature of 175°C to a temperature in °F

..... °F

(2)

(b) Change a temperature of 482°F to a temperature in °C

..... °C

(2)

(Total for question = 4 marks)

Edexcel IGCSE Mathematics A (4MA1) · Paper 1FR · June 2024 · Question 10

26 Solve the simultaneous equations

$$7x - 2y = 34$$
$$3x + 5y = -3$$

Show clear algebraic working.

$x =$ .....

$y =$ .....

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27 **12** Solve the simultaneous equations

$$4x + 3y = 9.6$$
$$6x + 5y = 16.8$$

Show clear algebraic working.

$x =$ .....

$y =$ .....

(Total for Question 12 is 4 marks)

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28 **6** Alison buys 5 apples and 3 pears for a total cost of $1.96
Greg buys 3 apples and 2 pears for a total cost of $1.22

Michael buys 10 apples and 10 pears.

Work out how much Michael pays for his 10 apples and 10 pears.
Show your working clearly.

$ .....

(Total for Question 6 is 5 marks)

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TOTAL FOR PAPER IS 96 MARKS